Video Transcript

Video 11: Introduction to Gaussian Naïve Bayes Theorem (05:22)

In this exercise, we're going to focus on using Gaussian Naive Bayes. The difference we have now is that we have distributions of our features. So, here we're going to use the height, weight, and foot size of a person to determine whether they're male or female. So, here's our training data. We've got four measurements for male and four for female. We're going to assume Gaussian distributions, so this is why we call it Gaussian Naive Bayes. And so, we're going to have for height, we're going to have one distribution for the males and a different distribution for the females.

So, you can see just by inspection of the data that the mean for the males is going to be different from the mean for the females for both height, weight, and foot size. Now, what we're going to rely upon for calculating the posterior probability is we're going to use the likelihood of seeing, for example, a person's weight. Let's suppose we have the person and they have a weight of 114 lbs. Then we'd use the Gaussian distribution that we have. This one is close for a female, but the male would be different to get us a probability. And here, we're getting a value of 0.04.

So, we take the value and this gives us the likelihood of seeing that value based on this probability density function on this Gaussian distribution. So, we can easily calculate the Gaussian distributions because all we need is the mean and the variance. So, step one is to write code to calculate the mean and the variance of each of the six Gaussian distributions. So, the six Gaussian distributions are for male. We have a distribution for height, so just those four values. We need the mean and the variance. Then for weight, again, mean and variance and for foot size. And then we have the same for the female.

So, we have a total of six Gaussian distributions that you're going to calculate. Now, here's our test data. We've got an unknown person. We don't know if they're male or female, but they've got a height of six foot, a weight of 130, and a foot size of eight. So, what you need to do is now to calculate each of the likelihoods. So here, this is the likelihood of seeing a certain height for

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either a male or female. So, X we have to substitute either the male Gaussian or the female Gaussian. Then what we can do is, as we saw before, because the heights and weights and foot sizes are assumed to be totally independent, then we can just multiply these probabilities.

So, this is, given that this is a female, this is the probability of seeing this height. So, it would be six, the height would be six here. And then the probability of seeing the weight of this person if it's female and we can compare that with the same probabilities for whether it's a male. And then comparing these two posterior, we can decide whether it's more likely that this person is a male or female. So, I'm going to give you sample code here for calculating each one of these. So, given that it's a female distribution and we see this height, what's the probability density function that we should insert here? So, here's the code.

So, this takes a mean and a variance. So, you need to plug in the ones that you've calculated. So, you've got all six of those to plug in. And then this would be the feature that you're looking at. And this will give you the probability of seeing that feature, the likelihood, in this case, I'm saying what's the likelihood of the person being 5 foot 8 or 5.8 feet when we have this Gaussian distribution? So, you need to calculate six of these to plug into these formula here. So, then you can calculate the posterior probabilities for male or female and tell me the likely gender for this person. Good luck. Bye for now.

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